Mohammad Zuhair Khan

Master's Student in Engineering Physics at Aalto University

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Profile

- Quantum Hardware
- Quantum Software
- Science Communicator
- Research-to-impact

WORK EXPERIENCE

Data Engineer
- Build the data and automation layer for an early-stage venture-capital fund on a new AWS data warehouse.
- Quarter-toggleable fund reporting that reproduces every reported figure, and performance-vs-plan dashboards built from the data lake.
- Slack bot that reviews investment documents against the firm's templates and flags deviations.

Jun 2026 - Present
Lifeline Ventures

Trainee
Harbor by AaltoES
- Selected through Aaltoes' competitive Harbor programme for a project engagement.
- Automated quarterly-review reporting and built a fund-level performance-vs-plan modelling tool.
- Mapped fragmented investment data toward a unified model and scoped AI agents for surfacing investment insights.

May 2026 - Jun 2026
Lifeline Ventures

Research Trainee
- Benchmarked and helped launch the VTT Q50 quantum computer.
- Optimised GHZ state preparation using graph theory and qubit characterisation.

Jun 2024 - Dec 2025
Quantum Algorithms & Software Team, VTT

Research Assistant
Bachelor's Thesis
- Python-based data analysis for experimental measurements.
- Operated dilution refrigerators and Vector Network Analysers (VNAs).
- Used the Triton cluster for FEM simulations.

Jun 2023 - May 2024
Pico Group, Aalto University

EDUCATION

M.Sc in Engineering Physics
Recipient of Category A scholarship (100% waiver) and Professor E. J. Nyström scholarship

Aug 2024 - Present
Aalto University

B.Sc in Quantum Technology
Graduated with Honours, with a minor in Computer Science
Recipient of Category A scholarship (100% waiver), and Dean's incentive scholarship (2022 - 2024)

Aug 2021 - Jun 2024
Aalto University

ABOUT ME

Interested in quantum technology, agentic AI, mathematical optimisation and all things deep tech.

" In science, there are no shortcuts to truth. "

LANGUAGES

Bengali	*C2**	●●●●●● *Native*
English	*C1**	●●●●●○
Finnish	*A1**	●○○○○○

*Using CEFR rating for language proficency.

Python	●●●●●○
LaTeX	●●●●●○
Scala	●●●○○○
MATLAB	●●●○○○
SQL	●●●○○○
R	●●○○○○

SKILL SET

- Quantum hardware (QPUs, dilution refrigerators)
- Quantum SDKs (Qiskit, Cirq, Q#, Pennylane, Bloqade, QuTiP)
- Control Software (AWGs, VNAs)
- Multiphysics Simulations (COMSOL)
- Version Control (Git)
- HPC & scientific computing (Triton cluster)

PUBLICATIONS

B.Sc Thesis	**Designing a 3D Cavity for Photon-Number Cooling Using a Quantum Circuit Refrigerator** Aalto University, Pico Group, 2024. Supervised by Matti Raasakka, advised by Shuji Nakamura. Aluminium and copper 3D cavity resonators coupled to a quantum circuit refrigerator, measured by VNA at room and millikelvin temperatures and compared against finite-element simulations.

TALKS & PRESENTATIONS

ENCCS NordIQuEst Quantum Autumn School 2024, KTH	**Introduction to Helmi** — Invited Speaker Introduced Finland's Q5 quantum computer and delivered a hands-on tutorial on GHZ state fidelity estimation, readout error mitigation, and Multiple Quantum Coherences on real quantum hardware. Recording Slides Notebook Dec 2024 · Stockholm, Sweden
Roosh Circle Papers Club	**Quantum ML Battle** — Invited Speaker A debate on near-term versus fault-tolerant quantum machine learning. Presented the fault-tolerant case: how it compares with near-term approaches, and what it would take to bring its practical use forward. May 2024 · Online

LEADERSHIP & OUTREACH

Vice Chair	• Vice Chair of AaltoAI, an AI-focused student community at Aalto University.	Jun 2026 - Present AaltoAI
Founder & Organiser	• Founded and single-handedly run the Aalto Quantum Journal Club, a student-focused reading group at Aalto University exploring quantum technology fundamentals; curate paper selections, facilitate discussions, and grow community participation.	Oct 2024 - Present Aalto Quantum Journal Club

SELECTED PROJECTS

Pause	A translucent floating break-timer for Android that counts down over whatever app you are in, then eases you into a full-screen 4-7-8 breathing wind-down. Shipped with tests and CI. Repository
QASM toolkit	Estimates a peaked circuit's peak bitstring, simulates such circuits with matrix product states, and optimises them with equivalence checking. Repository
Agentic GraphRAG	Retrieval over live Finnish tax regulation, reranked on a typed graph of legal relationships; cross-lingual retrieval lifted similarity to 0.861 over 121k chunks. Repository

AWARDS & ACHIEVEMENTS

Mimir	Deep Dive Commercialisation Competition — **1st Place** Won the multi-round competition with a business strategy for a research-stage invention, developed with **Willo Technologies**.
Dash	Dash Hackathon 2023: Sumitomo SHI FW Challenge — **2nd Place**
Junction	Quantum Hack 2026: QMill Challenge — **Top 5** by audience vote, presented to the jury
Womanium	Quantum Hackathon 2022: IQM DAQC-VQE Challenge — **Finalist**
QCoder	QCoder Programming Contest 006 — **15th**
IMC	Prosperity 3 Algorithmic Trading Competition — **25th in Italy**

CERTIFICATIONS

IBM	Basics of Quantum Information, Variational Algorithm Design, Quantum Business Foundations, Practical Introduction to Quantum-Safe Cryptography (2024). Quantum Challenge Fall 2022 & Spring 2023 — **Advanced**.
Other	QWorld QSilver 14 & QBronze 96 (2022). Qubit by Qubit Quantum Winter School, Microsoft (2023). Introduction to Programming with Neutral Atoms, QuEra.